🏀 Smart Sports Team Performance Analyzer using Clustering

1. Introduction

Sports analytics has become an important area where data analysis and machine learning are used to evaluate team performance and improve strategies. By analyzing team statistics such as matches won, points scored, defense performance, and possession percentage, we can understand how teams perform compared to others.

This project focuses on analyzing sports teams using machine learning clustering techniques. The system groups teams based on their performance statistics and classifies them into strong, average, or weak teams. This helps in understanding team strengths and comparing team performances effectively.

The project demonstrates the use of unsupervised learning techniques in machine learning to analyze sports data.

2. Objective

The main objectives of this project are:

- To analyse team performance using statistical data.
- To group teams based on similar performance characteristics.
- To visualize team clusters using graphs.
- To create an interactive dashboard for team analysis.
- To demonstrate machine learning techniques in sports analytics.

3. Dataset Description

The dataset contains information about different sports teams and their performance statistics.

Attributes used in the dataset:

Attribute	Description
Team	Name of the team
Matches Won	Number of matches won by the team
Matches Lost	Number of matches lost
Points Scored	Total points scored by the team
Defense Score	Defensive strength rating
Possession	Ball possession percentage

Attribute	Description
Rebounds	Number of rebounds
Fouls	Number of fouls committed

Example dataset entry:

Team	Matches Won	Points Scored	Defense Score	Possession
Lakers	18	2100	85	60
Warriors	15	1950	80	58
Bulls	8	1400	65	50

4. Methodology

The system follows several steps to analyze team performance.

1. Data Collection

The dataset is stored in a CSV file containing team statistics.

2. Data Preprocessing

Data is cleaned and prepared for analysis. Features are normalized using scaling techniques to ensure fair comparison.

3. Feature Selection

Important features used for clustering include:

- Matches Won
- Matches Lost
- Points Scored
- Defense Score
- Possession
- Rebounds
- Fouls

4. Clustering

The teams are grouped using **Hierarchical Clustering** to identify similar performance patterns.

5. Dimensionality Reduction

To visualize clusters clearly, **Principal Component Analysis** is used to reduce the dataset to two dimensions.

6. Visualization

Graphs are used to visualize team clusters and performance comparisons.

5. Tools and Technologies Used

Tool	Purpose
Python	Programming language
Streamlit	Building the interactive dashboard
Pandas	Data processing
Scikit-learn	Clustering algorithms
Matplotlib	Data visualization

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6. System Architecture

The workflow of the system is shown below:

Dataset

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Data Preprocessing

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Feature Selection

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Hierarchical Clustering

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PCA Visualization

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Performance Analysis

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Dashboard Output

7. Implementation

The project is implemented using Python and Streamlit. The dataset is loaded and processed using Pandas. Clustering is performed using hierarchical clustering techniques to group teams based on performance. PCA is used to visualize the clusters.

The Streamlit interface allows users to:

- View team statistics
- Select a team and see details
- Compare teams
- View team rankings
- Analyze cluster graphs

8. Results

The system successfully groups teams based on their performance metrics.

Example cluster classification:

Cluster	Team Category
Cluster 0	Strong Teams
Cluster 1	Average Teams
Cluster 2	Weak Teams

The visualization graphs clearly show how teams are grouped according to their performance.

9. Applications

This system can be applied in several areas such as:

- Sports analytics platforms
- Team performance analysis
- Coaching strategy planning
- Player and team comparison tools
- Sports data research

10. Advantages

- Easy to analyze team performance
- Visual representation of team clusters
- Interactive dashboard for analysis
- Useful for sports analytics

11. Limitations

- Dataset size is small
- Real-world sports data may require more features
- Model accuracy depends on the quality of input data

12**. Conclusion**

This project demonstrates how machine learning techniques can be applied to analyze sports team performance. By using hierarchical clustering and PCA, teams can be grouped based on their statistical performance.

The developed system provides an interactive platform for analyzing sports data and understanding team strengths and weaknesses. This project highlights the importance of data-driven decision making in modern sports analytics.